Economics S5040: Dynamic Stochastic Modeling
Lecture 7 — Introduction to Dynamic Programming

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1 Why Dynamic Programming?

In an earlier lecture we studied saving and investment over a small number of periods. The essential difficulty was that a decision made today changes the opportunities available tomorrow.

The general principle is simple: *What we choose now changes our future irrevocably.* Dynamic programming formalizes this idea. It gives us a way to solve problems in which current choices affect future opportunities.

2 The Five Ingredients

A dynamic program has five basic elements:

1. a **state space** X;
2. a **control set** U;
3. a **reward function** $r(x, u)$;
4. a **transition rule** describing the distribution of next period's state;
5. a **discount factor** $\beta \in [0, 1)$.

At date t, the state x_t summarizes the information relevant for the decision. The control u_t is the action chosen by the agent.

In a stochastic problem we can write the transition rule schematically as

$$x_{t+1} \sim g(x_t, u_t).$$

It is important to emphasize that this transition rule is *random*. You can make very good choices about your education and life, but this classroom could be hit by a meteorite in the next few minutes, and your best laid plans will be obliterated.

Given x_0–the state in which we find outselves right now–the agent's problem is to choose a sequence of actions

$$\max_{\{u_t\}} E\left[\sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \,\middle|\, x_0\right]$$

subject to

$$u_t \in U, \qquad x_{t+1} \sim g(x_t, u_t).$$

The notation is compact, but the economics is familiar: choose what to do today, while recognizing that today's choice changes tomorrow's possibilities.

3 What Does It Mean to Solve the Problem?

In a stochastic environment, a solution cannot simply be a predetermined sequence

$$(u_0, u_1, u_2, \ldots),$$

because we do not yet know which future states will occur.

Instead, we need a plan for every possible contingency.

A **policy function**

$$h : X \to U$$

specifies the action to take in every state:

$$u_t = h(x_t).$$

Thus a policy is a contingent plan.

If h is optimal, the associated **value function** is

$$V(x_0) = E\left[\sum_{t=0}^{\infty} \beta^t r(x_t, h(x_t)) \,\middle|\, x_0\right].$$

The value function answers the question:

> If I begin in state x_0 and behave optimally from now on, how valuable is that position?

It is important to distinguish the two objects:

policy function: what should I do, come what may?

and

value function: how good is it to be here right now, knowing that my future plans are as good as can be?

4 The Bellman Equation

Suppose we knew the value function. If the current state is x and we choose control u, the value of that choice is

$$r(x, u) + \beta E[V(x')].$$

The first term is the reward today. The second is the discounted expected value of the state we reach tomorrow. It is common notation in macroeconomics to let x' denote next periods state. Notice that $x' \sim g(x, u)$ depends upon what we are choosing today. So the second term depends upon x, u and the shape of $V(\cdot)$.

The optimal control maximizes this expression. Therefore the value function must satisfy

$$\boxed{V(x) = \max_{u \in U} \left\{ r(x, u) + \beta E[V(x')] \right\}.}$$

This is the **Bellman equation**. The problem with this expression is that the value function appears on both sides of the equation!

It is the central equation of dynamic programming.

The Bellman equation divides the problem into

$$\underbrace{r(x,u)}_{\text{reward today}} + \underbrace{\beta E[V(x') \mid (x,u)]}_{\text{value of the future}}.$$

The control can matter in both places: it can affect today's reward and the distribution of tomorrow's state.

5 Optimal Growth as a Dynamic Program

Consider an infinite-horizon growth problem. Capital today is k_t and output is

$$f(k_t) = Ak_t^\alpha, \qquad 0 < \alpha < 1.$$

Output can be consumed or carried into the next period as capital:

$$c_t + k_{t+1} = Ak_t^\alpha.$$

Preferences are

$$\sum_{t=0}^{\infty} \beta^t \log c_t.$$

Following the notation of the original notebook, let the control u_t be next period's capital:

$$u_t = k_{t+1}.$$

Then consumption is

$$c_t = Ak_t^\alpha - u_t.$$

The five ingredients are therefore

state:	k_t,
control:	$u_t = k_{t+1}$,
reward:	$r(k_t, u_t) = \log(Ak_t^\alpha - u_t)$,
transition:	$k_{t+1} = u_t$,
discount factor:	β.

There is no uncertainty in this example. This aspect of the problem makes it a lot easier to solve analytically.

The Bellman equation is

$$\boxed{V(k) = \max_u \left\{\log(Ak^\alpha - u) + \beta V(u)\right\}.}$$

Another special fact about this problem is that next period's state k' does not depend explicitly on the current state k.

6 Guess and Verify

For this special problem, we can solve the Bellman equation analytically.

Guess

$$V(k) = E + F \log k$$

for constants E and F. The technique we are learning is called the method of undetermined coefficients. Still, we had to guess a shape for the value function.

If the guess is correct, the first-order condition for u is

$$-\frac{1}{Ak^\alpha - u} + \beta V'(u) = 0.$$

Since

$$V'(u) = \frac{F}{u},$$

we obtain

$$-\frac{1}{Ak^\alpha - u} + \beta \frac{F}{u} = 0.$$

Solving,

$$u = \frac{\beta F}{1 + \beta F} Ak^\alpha.$$

Thus our guess implies that the optimal policy saves a constant fraction of output.

Now substitute the implied policy back into the Bellman equation. Matching the coefficient on $\log k$ gives

$$F = \alpha + \alpha\beta F.$$

Hence

$$\boxed{F = \frac{\alpha}{1 - \alpha\beta}.}$$

Therefore

$$\frac{\beta F}{1 + \beta F} = \alpha\beta,$$

and the optimal policy is

$$\boxed{k_{t+1} = \alpha\beta A k_t^\alpha.}$$

The household saves the constant fraction

$$\boxed{\alpha\beta}$$

of current output.

The saving rate is larger when capital's share is larger and when the household is more patient.

The algebra of solving for the coefficients E and F

This subsection is optional.

We guessed that the value function has the form

$$V(k) = E + F \log k.$$

From the first-order condition, we found that the optimal control must satisfy

$$u = \frac{\beta F}{1 + \beta F} A k^{\alpha}.$$

Define

$$s \equiv \frac{\beta F}{1 + \beta F}.$$

Then the implied policy is

$$u = s A k^{\alpha}.$$

Since

$$c = A k^{\alpha} - u,$$

optimal consumption is

$$c = (1 - s) A k^{\alpha}.$$

Now substitute the implied policy back into the Bellman equation:

$$V(k) = \log(A k^{\alpha} - u) + \beta V(u).$$

Using

$$u = s A k^{\alpha}$$

and our guess

$$V(k) = E + F \log k,$$

we obtain

$$E + F \log k = \log \left[(1 - s) A k^{\alpha} \right] + \beta \left[E + F \log \left(s A k^{\alpha} \right) \right].$$

Expand the logarithms:

$$\log \left[(1 - s) A k^{\alpha} \right] = \log(1 - s) + \log A + \alpha \log k$$

and

$$\log \left(s A k^{\alpha} \right) = \log s + \log A + \alpha \log k.$$

Hence

$$\begin{aligned} E + F \log k = {} & \log(1 - s) + \log A + \alpha \log k \\ & + \beta E + \beta F \log s + \beta F \log A + \alpha \beta F \log k. \end{aligned}$$

Since this equality must hold for every value of k, the coefficients on $\log k$ must be equal. The coefficient on $\log k$ on the left-hand side is

$$F,$$

while the coefficient on the right-hand side is

$$\alpha + \alpha \beta F.$$

Therefore

$$F = \alpha + \alpha \beta F.$$

Rearranging,

$$F - \alpha \beta F = \alpha,$$

so

$$F(1 - \alpha\beta) = \alpha.$$

Hence

$$\boxed{F = \frac{\alpha}{1 - \alpha\beta}.}$$

We can also solve for the constant E. Matching the terms that do not depend on k gives

$$E = \log(1 - s) + \log A + \beta E + \beta F \log s + \beta F \log A.$$

Therefore

$$(1 - \beta)E = \log(1 - s) + (1 + \beta F) \log A + \beta F \log s.$$

Hence

$$\boxed{E = \frac{\log(1 - s) + (1 + \beta F) \log A + \beta F \log s}{1 - \beta}.}$$

Using

$$s = \alpha\beta$$

and

$$F = \frac{\alpha}{1 - \alpha\beta},$$

this can be written entirely in terms of the primitive parameters A, α, and β.

In most economic applications, the coefficient F and the implied policy function are more important than the constant E. The constant E affects the level of the value function, but not the optimal saving rule.

7 Our First Euler Equation

The first-order condition we derived is important enough to pause over:

$$-\frac{1}{Ak_t^\alpha - k_{t+1}} + \beta V'(k_{t+1}) = 0.$$

Now I have written it with explicit time subscripts. Since

$$c_t = Ak_t^\alpha - k_{t+1},$$

we can write this as

$$\frac{1}{c_t} = \beta V'(k_{t+1}).$$

This is the first appearance in this course of an **Euler equation**. To see its familiar economic interpretation, consider the envelope condition.

The Bellman equation is

$$V(k_t) = \max_{k_{t+1}} \left\{ u(c_t) + \beta V(k_{t+1}) \right\},$$

where

$$c_t = f(k_t) - k_{t+1}.$$

The envelope condition gives

$$V'(k_t) = u'(c_t)f'(k_t).$$

This expression answers the question, What is the value of having one more piece of capital today, if we know that we are always going to invest future savings optimally? The answer is the marginal utility of consumption multiplied by the marginal product of capital. The extra piece of capital will give us $f'(k_t)$ more GDP today, and we value that extra GDP according to the current marginal utility of consumption $u'(c_t)$.

One period ahead,

$$V'(k_{t+1}) = u'(c_{t+1})f'(k_{t+1}).$$

Substituting this expression into the first-order condition gives

$$\boxed{u'(c_t) = \beta u'(c_{t+1})f'(k_{t+1}).}$$

This is the **Euler equation** for the optimal growth problem. It is important to see when these expressions are being evaluated. All these quantities occur when one follows the best possible savings plan.

We can also rearrange the Euler equation as

$$\boxed{\frac{u'(c_t)}{\beta u'(c_{t+1})} = f'(k_{t+1}).}$$

The left-hand side is the household's marginal rate of substitution between consumption today and consumption tomorrow. We already know that quantity is the gross real interest rate from t to $t+1$. The right-hand side is the marginal return from transferring one additional unit of resources into capital. It is called the marginal efficiency of investment.

Thus the economic principle is the same one that we encountered in our two-period model:

$$\boxed{\text{MRS} = \text{intertemporal return} = \text{Gross real interest rate}}$$

What is new is the dynamic setting. This condition must hold at every date along an optimal path. Euler equations of this form will appear repeatedly throughout the rest of the course.

8 The Long-Run Evolution of Capital

Solving the dynamic program gives us a useful by-product: the law of motion of the economy when agents behave optimally:

$$k_{t+1} = \alpha\beta A k_t^{\alpha}.$$

Taking logarithms,

$$\log k_{t+1} = \log(\alpha\beta A) + \alpha \log k_t.$$

The steady state satisfies

$$k^* = \alpha\beta A (k^*)^{\alpha}.$$

Thus

$$\boxed{k^* = (\alpha\beta A)^{1/(1-\alpha)}.}$$

A simple Python loop can trace the optimal path:

```
def optgrow(A, alpha, beta, k0, T):
    path = [k0]
    for t in range(T):
        k1 = alpha * beta * A * k0**alpha
        path.append(k1)
        k0 = k1
    return path
```

The important point is not the particular formula. It is the sequence of ideas:

dynamic problem → Bellman equation → value function → policy function → law of motion.

9 What to Remember

The essential ideas from today are:

1. The **state** describes where the agent is.
2. The **control** describes what the agent chooses.
3. A **policy function** gives the action in every possible state.
4. The **value function** gives the value of beginning in each state and behaving optimally.
5. The Bellman equation is

$$V(x) = \max_u \{r(x,u) + \beta E[V(x') \mid (x,u)]\}.$$

6. In special cases we can solve a Bellman equation analytically by guessing the form of the value function and verifying the guess.

In the next lecture we turn to a small stochastic problem with finite state and control spaces. There we will see how to solve a Bellman equation when an analytic solution is unavailable.